Response to the reviewer (second round)

A reliability-constrained, multi-objective framework for minimum-mass design of steel frame structures under material and loading uncertainty

[Author name]

We thank the reviewer for a second careful report. We agree that a response letter is not evidence in itself. For each point below, we therefore name the new computation, the table or figure where its result appears, and the sentences of the manuscript that changed. All new results were produced by the seeded scripts in the repository, and the commit that produced them is given in the manuscript (Data and code availability). Section, table and figure numbers refer to the revised manuscript.

Summary of the main changes

1. The three-storey design was re-optimised twice. The first design imposes the targets on 95% lower confidence bounds; the second adds a margin informed by the full-model discrepancy. Both designs were validated against full second-order re-analysis (Section 4.6).
2. The paper now distinguishes three claims everywhere: (P) the point estimate reaches the target, (I) the interval contains it, and (L) the lower bound exceeds it. "Within" results are no longer described as meeting the target (Section 2.5).
3. A confidence-bounded formulation, RBDO-C95, was run for all seven frames. It is presented as the recommended formulation when a minimum reliability is to be demonstrated (Sections 2.6 and 4.6, Table 16).
4. The excluded part of the failure tail was tested with 3 000 additional full re-analyses and a residual-risk estimate. The statement that excluded samples "cannot fail" was removed (Section 3.3, Table 6).
5. The full texts of Papadrakakis et al. (2004) and Truong and Kim (2017) were read, and Table 1 and the novelty statement were revised accordingly. The full text of Foley et al. (2007) could not be accessed.
6. Model-uncertainty-aware designs (RBDO-C-MU) quantify the extra steel needed to keep the target when model factors are included (Table 17).
7. An analysis of the effect of the 1% drift bias of the B_1–B_2 surrogate on the drift reliability was added (Table 16).

Major comments

Comment 1 (three-storey design). *The three-storey RBDO-C design ($\hat{\beta} = 2.995$ [2.98, 3.01]) is accepted as "within", while the full model gives $\Delta\beta = -0.07$ [−0.17, −0.01]. Re-optimise with a margin or a stricter acceptance rule, report the extra mass, and revise the abstract and conclusions.*

Response. We did both, and validated the resulting designs in the full model. The results are in Section 4.6 and Table 7.

- *RBDO-C (8188 kg)*, the previous design. Full model: $\hat{\beta}_{ULS}^{full} = 2.94$ [2.84, 3.01]. It is no longer described as meeting the target anywhere in the paper.
- *RBDO-C95 (8282 kg, +1.15%)*, with the target imposed on the SAA lower confidence bound. Surrogate: 3.068 (lower bound 3.053), i.e. claim L. Full model: 3.01 [2.89, 3.10] with $\Delta\beta = -0.09$ $[-0.21, -0.01]$. The point estimate reaches the target, but the interval contains it (claim P, I). The discrepancy is therefore systematic for this frame.
- *RBDO-C95 with a margin (8408 kg, +2.69%).* The ULS target is raised to 3.21, i.e. by the upper end of the discrepancy interval. Because the catalogue is discrete, the design jumps to $\hat{\beta}_{ULS} = 3.62$ [3.58, 3.66]. Full model: 3.61 [3.45, 3.73], or 3.52 with the heavy-tailed residual of the excluded samples. This meets the target with 95% confidence on the sampled strata.

The steel needed to meet the ULS target with 95% confidence in the full model is therefore 2.69% above RBDO-C and 5.17% above DDO. The abstract, Discussion and Conclusions (items 1, 3 and 4) now state this, and they describe the three-storey result as a limitation of the surrogate. We could not isolate the cause of the discrepancy, and the manuscript says so rather than offering an unverified explanation.

Comment 2 (acceptance rule). *Distinguish the point estimate, the interval containing the target and the lower bound exceeding it; present the confidence-bounded formulation as a substantive alternative; consider making it the primary recommendation; use precise wording.*

Response.

- *Three claims.* Section 2.5 now defines the three claims (P), (I) and (L). It states that only (L) establishes the target with 95% confidence and that a "within" result is inconclusive. The former sentence that accepted "within" as meeting the target was deleted.
- *Tables.* Table 16 classifies every design by the strongest claim, and Table 13 marks (L) in bold and (P/I) in italic.
- *Confidence-bounded formulation.* RBDO-C95 is defined in Section 2.6 and was run for all seven frames. All its designs meet both targets with 95% confidence in independent verification. It costs 1.15–1.55% more steel than RBDO-C in three frames and nothing in three. In the 6-m bay frame it found a design 0.26% lighter than the RBDO-C runs; we report this as a limitation of the search budget. Against DDO, RBDO-C95 costs 1.1–11.7%.
- *Recommendation.* The Discussion and Conclusion 4 now recommend RBDO-C95 whenever a minimum reliability is to be demonstrated. RBDO-C is kept for the like-for-like comparison with DDO, which is also judged on nominal values.
- *Wording.* The abstract and conclusions now state the claims separately: six of seven RBDO-C designs reach (L), and all RBDO-C95 designs do.
- *Drift limits.* The comparison with tightened drift limits was redone for both claim levels (Section 4.4). Against deterministic designs with claim (L), RBDO-C95 is lighter in three frames and 0.27% (high wind) and 1.55% (base) heavier in two.

Comment 3 (excluded tail). *An observed maximum ratio of 1.049 is not a bound. Validate the threshold or estimate the residual risk, explain the estimator, and check a lower threshold or a larger sample.*

Response. We agree, and the sentence that samples below 0.85 "cannot fail" has been removed. The estimator is now described as resting on an assumption that is tested (Section 3.3):

- *Lower threshold.* For every validated design, 300 samples of the stratum $0.70 \leq U_s < 0.85$ of the same pool were re-analysed with the full model, 3 000 analyses in total. None failed. The largest observed ratio is now described as an observed maximum: 1.060 in this stratum and 1.061 over all 13 042 re-analysed samples. Lowering the threshold to 0.70 leaves the point estimates unchanged.
- *Limits of this check.* The 95% intervals become much wider (lower ends 0.35–0.69 below the point estimates). This is because the stratum carries 6–37% of the pool probability, and 300 failure-free samples cannot bound a probability of order 10^{-3} in it. We state this explicitly; a distribution-free bound would need tens of thousands of full analyses per design.
- *Residual-risk estimate.* For all unanalysed samples with $U_s < 0.85$, the residual is estimated from a fitted model of $\ln(U_{full}/U_s)$: normal, and Student t_3 as a heavy-tailed alternative. The normal model gives $< 10^{-23}$. The t_3 model gives 2.7×10^{-5} to 1.8×10^{-4}, which lowers $\hat{\beta}^{full}$ by 0.01–0.02 for most designs, by 0.06 for nine storeys and by 0.15 for 6-m bays. No target classification changes, except that the RBDO-C95 three-storey design drops from 3.01 to 2.97.

All of this is in Table 6. The summary paragraph of Section 3.3 and Conclusion 1 now say that the validation is statistical and that the tail beyond the observed maximum is modelled, not observed.

Comment 4 (literature). *Read the full texts of Papadrakakis et al. (2004), Foley et al. (2007) and Truong and Kim (2017); check the four claimed contributions; update Table 1.*

Response.

- *Papadrakakis et al. (2004)*, full text read (the openly available PDF in the BazTech repository). Two of our earlier implicit claims did not survive this reading, and we have corrected them. They imposed the Eurocode 3/8 checks in both the deterministic and the reliability-based problem, and they compared the two optima. The reliability-based design was heavier: their text says about 26%, while the volumes in their Table 6 differ by 34%. Their probabilistic constraint is on interstorey drift $> 4\%$ of storey height under seismic loading, estimated with 5000 (MCS) or 1000 (LHS) samples, without confidence intervals. The introduction now states that imposing code checks alongside a reliability constraint, and finding that the reliability-based design is heavier, are *not* new.
- *Truong and Kim (2017)*, full text read (an openly accessible copy of the article). Their RBDO of steel space frames under gravity and wind also keeps deterministic strength, $H/400$ drift and constructability constraints. Their failure criterion is load-carrying capacity from plastic-hinge advanced analysis. Drift is not a probabilistic constraint. The sampling estimator is verified against crude MCS on benchmark problems, but the failure probabilities of the final frame designs are not re-estimated independently, and no confidence intervals are given.
- *Foley et al. (2007).* We could not access the full text: the publisher's page shows only the abstract, and no open copy was found. We say so plainly. Its row in Table 1 and the rows of the other studies we could assess only from abstracts are marked with †. The text makes no novelty claim relative to them.
- *Revised contribution.* It is now stated relative to the two studies read in full. In neither did we find: (i) a comparison under a code-calibrated gravity–wind model with deterministic optima, including tightened drift limits and RBDO with and without code checks; (ii) a probabilistic drift

constraint next to the ULS constraint; (iii) confidence intervals or independent re-estimation for the final designs; (iv) a tail validation of the surrogate against full second-order re-analysis; or (v) a code release. Table 1 was updated with the full-text information.

Comment 5 (model uncertainty). *Make the conditional nature of the baseline explicit; avoid suggesting a real-world target is met; explain the purpose of the model-factor analysis; if feasible, give a model-uncertainty-aware design.*

Response.

- *Conditional statements.* The conditional nature of the indices is now stated in Section 2.5 (last paragraph), in Section 4.8, at the start of the Conclusions and in the abstract. Section 4.8 says explicitly that the baseline designs do *not* meet β_T once model uncertainty with c.o.v. 0.10 is counted.
- *Purpose of the model factors.* Section 2.4 explains that they are used both as a sensitivity study and, for two frames, inside the constraints.
- *Model-uncertainty-aware designs.* RBDO-C-MU (c.o.v. 0.05) was run for the base and three-storey frames (Table 17). Keeping $\beta_T = 3.0$ with model uncertainty included costs +1.34% (base, $\hat{\beta}_{ULS} = 3.047$ with the factors) and +2.69% (three storeys) relative to RBDO-C. Without the factors, the RBDO-C designs have only 2.77 and 2.70. Designs for c.o.v. 0.10 were not computed; the manuscript says they would be heavier.

Comment 6 (cost model). *Keep economy tied to mass or the illustrative index; do not imply validation against project costs; state how the penalty ranges relate to plausible costs or that they are hypothetical.*

Response. Section 2.7 now states that τ and κ are hypothetical sensitivity parameters, not estimates of fabrication or connection costs, and that the ranking has not been checked against project costs. The Discussion and Conclusion 5 limit all economic statements to steel mass and the uncalibrated index.

Additional technical and editorial comments

Comment Table 3 vs. response letter (fabrication factor).

Response. Our first response letter was wrong on this point. The fabrication-factor model (normal, mean 1.00, c.o.v. 0.05) is taken from Ellingwood et al. (1980), as Table 3 states, and it is sourced, not assumed. The assumed parameters are the mean-to-nominal ratio of F_y (1.10), its inter-member correlation (0.5), the out-of-plumbness standard deviation and the model factors. This is what the "Status" column of Table 3 shows.

Comment B_1–B_2 drift bias near the drift target.

Response. A new computation increases all surrogate drifts by 1% and re-evaluates β_{drift} with 10^6 samples (Table 16, last columns; stage `driftbias`). This lowers β_{drift} by 0.026–0.028. The base and high-wind RBDO-C designs (1.507, 1.511) would then fall below 1.5, and the nine-storey design would become inconclusive. The RBDO-C95 designs keep claim L, except the nine-storey design, which is identical to its RBDO-C design. Section 4.6 states that drift targets met by less than about 0.03 are met only to the accuracy of the B_1–B_2 approximation.

Comment Seeds and robustness.

Response. Sections 2.6 and 4.1 and the "demonstrated" paragraph of the Discussion describe the evidence as robustness within the tested search budget. The lighter 6-m bay design found by the RBDO-C95 run is reported as an example of the limits of that budget.

Comment Fully correlated loads and common imperfection.

Response. A paragraph "Dependence and spatial variability" was added to Section 2.4. It states these assumptions and warns against extrapolating to buildings with substantial spatial variability. The "not demonstrated" list in the Discussion repeats this.

Comment Identifiable commit and executable workflow.

Response. The Data and code availability statement now names the commit that produced the results of this version. `reproduce.sh` and `jobs.txt` include every new run and stage (confidence-bounded and model-uncertainty designs, `validateC`, `driftbias`). A release tag and DOI will be created on acceptance.

Comment Author placeholders.

Response. The author names and affiliations remain placeholders in this version. They will be completed before formal submission.

Comment Headline claims.

Response. The abstract and Conclusions were rewritten. They state the three claim levels separately, give the three-storey full-model result, and state that all indices are conditional on the models.